Assignment-4.5

“Network Testing”

TASK-1.Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other..On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

Result

I used the ping command in Command Prompt to test communication between my laptop and another device with the IP address 10.230.50.87.
The test was successful. All four packets were sent and received, with 0% packet loss. The minimum response time was 7 ms, the maximum was 27 ms, and the average was 13 ms.
Conclusion

The ping test showed that my laptop could communicate with the device at 10.230.50.87. Since all four packets were received successfully, there was no packet loss during this test. This indicates that the connection between the two devices was working at that time.

TASK-2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

To use the `tracert` command to identify the network route from my laptop to YouTube and find the number of hops.

Procedure

1. I opened Command Prompt on my Windows laptop.
2. I entered the command `tracert www.youtube.com`.
3. I pressed Enter and waited for the command to finish.
4. The command displayed the route from my laptop to YouTube, showing the hops along the way.

Result

The traceroute command successfully reached YouTube's destination IP address, `2001:4860:482c:400::`, in **10 hops**.

- **Total hops:** 10
- **First hop:** Responded in 4–5 milliseconds.
- **Final hop:** Reached the destination in 33–84 milliseconds across the three probes.

- **Timed-out hops:** Hops 2, 4, 5, 7, 8, and 9 did not respond to the traceroute requests.



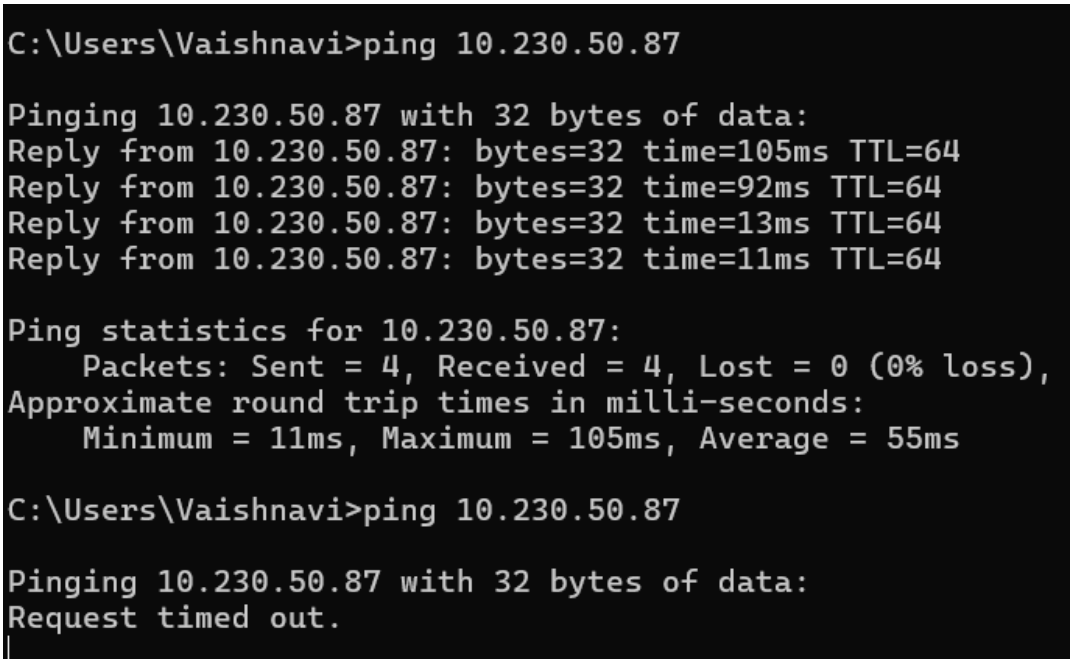
TASK-3.Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Procedure

1. I connected my laptop and my friend's phone to the same network using a mobile hotspot.
2. I opened Command Prompt on my laptop.
3. I entered the following command to test connectivity:

   ```
   ping 10.230.50.87
   ```

4. Initially, the ping test was successful. All four packets were received, with 0% packet loss.
5. I disconnected my friend's phone from the Wi-Fi network.
6. I ran the same ping command again to check whether the phone was still responding.



TASK-4 Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

To view the IP addresses and MAC addresses of devices detected by my laptop on the network using the `arp -a` command.

Procedure

1. I connected my laptop to my Samsung phone's mobile hotspot.
2. I opened Command Prompt on my laptop.
3. I typed `arp -a` and pressed Enter.
4. The command displayed the IP addresses, physical (MAC) addresses, and entry types of devices known to my laptop.
5. I examined the results to identify the devices.

Table of Network Devices

No.	IP Address	MAC Address	Device
1	10.230.50.149	Not displayed in this output	My laptop
2	10.230.50.87	6e-e4-75-53-db-62	Another device, possibly my friend's phone
3	10.230.50.116	ae-c4-68-95-01-34	Samsung phone's hotspot gateway

Note: The device at `10.230.50.87` responded to my earlier ping test, but its identity cannot be confirmed from the ARP output alone.

Understanding the Results

- **IP Address:** Identifies a device on the local network.
- **MAC Address:** Identifies a network interface at the local network level.
- **Dynamic:** An entry learned automatically through network communication.
- **Static:** An entry used for special addresses, such as broadcast and multicast addresses.

The other interfaces (`192.168.153.1` and `192.168.76.1`) belong to additional network interfaces on my laptop. Their presence does not necessarily mean that additional physical devices are connected to my phone's hotspot.

Result

The `arp -a` command displayed two dynamic entries on my laptop's active Wi-Fi network: `10.230.50.87` and `10.230.50.116`. These entries showed the IP and MAC addresses of devices known to my laptop.

TASK-5 Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Include screenshots or command outputs if possible.

Objective

To connect devices to a network and test connectivity using the `ping`, `tracert`, and `arp` commands.

1. Network Setup

I connected my laptop to my Samsung Android phone's mobile hotspot. I used the `ipconfig` command in Command Prompt to check my network details.

Network Setting	Value
IPv4 Address	10.230.50.149
Subnet Mask	255.255.255.0
Default Gateway	10.230.50.116

```
   Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40c1:1010:fa7:355f:7cf3:9093:f8c3
   Temporary IPv6 Address. . . . . . : 2409:40c1:1010:fa7:b1f6:8b88:e9a2:f6a7
   Link-local IPv6 Address . . . . . : fe80::549b:6781:cfe9:dbcf%16
   IPv4 Address. . . . . . . . . . . : 10.230.50.149
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::acc4:68ff:fe95:134%16
                                       10.230.50.116
```

2. Testing Connectivity Using Ping

I used the command `ping 10.230.50.87` to check communication between my laptop and another device.

Initially, the test was successful. All four packets were received, with 0% packet loss. The average response time was 13 milliseconds. This showed that the device was responding to my laptop's ping requests.

```
Pinging 10.230.50.87 with 32 bytes of data:
Reply from 10.230.50.87: bytes=32 time=27ms TTL=64
Reply from 10.230.50.87: bytes=32 time=10ms TTL=64
Reply from 10.230.50.87: bytes=32 time=9ms TTL=64
Reply from 10.230.50.87: bytes=32 time=7ms TTL=64

Ping statistics for 10.230.50.87:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 27ms, Average = 13ms
```

3. Tracing the Route to YouTube

I used the command `tracert www.youtube.com` to trace the network route to YouTube.

The command successfully reached YouTube in **10 hops**. Some intermediate hops did not respond and displayed asterisks. This can happen when network devices do not respond to traceroute requests.

```
C:\Users\Vaishnavi>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482c:400::]
over a maximum of 30 hops:

  1     5 ms     5 ms     4 ms  2409:40c1:1010:fa7::d6
  2     *        *        *     Request timed out.
  3    59 ms    52 ms    27 ms  2405:200:5210:2:3925::1
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    73 ms    29 ms    30 ms  2405:200:801:2d00::24b
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    84 ms    37 ms    33 ms  2001:4860:482c:400::

Trace complete.
```

4. Checking Network Devices Using ARP

I used the command `arp -a` to view the IP and MAC addresses known to my laptop.

IP Address	MAC Address	Device
10.230.50.87	6e-e4-75-53-db-62	Another device; exact identity not confirmed
10.230.50.116	ae-c4-68-95-01-34	Hotspot gateway

The ARP command helped me view network address information for devices known to my laptop. It may not display every connected device.

```
C:\Users\Vaishnavi>arp -a

Interface: 10.230.50.149 --- 0x10
  Internet Address      Physical Address      Type
  10.230.50.87          6e-e4-75-53-db-62     dynamic
  10.230.50.116         ae-c4-68-95-01-34     dynamic
  10.230.50.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.153.1 --- 0x12
  Internet Address      Physical Address      Type
  192.168.153.254       00-50-56-f6-74-7d     dynamic
  192.168.153.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.76.1 --- 0x16
  Internet Address      Physical Address      Type
  192.168.76.254        00-50-56-f3-7c-51     dynamic
  192.168.76.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

5. Testing a Disconnected Device

After disconnecting the device with IP address `10.230.50.87`, I ran the same ping command again.

This time, all four requests timed out. The result showed four packets sent, zero received, and 100% packet loss. This indicated that the device was not responding to my laptop at that time.

```
C:\Users\Vaishnavi>ping 10.230.50.87

Pinging 10.230.50.87 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.230.50.87:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Conclusion

This practical helped me understand basic network testing and troubleshooting. The ping command tested communication between devices, the tracert command showed the route to YouTube, and the ARP command displayed IP and MAC addresses known to my laptop. I also observed that disconnecting a device can cause ping requests to time out.